SPORTING LONG GUNS

By the middle of the 16th century, some sporting guns had developed "rifled barrels" in which parallel spiral grooves were cut along the bore of the barrel. Firing these "rifles" imparted a spin to the round lead balls used as ammunition. This rotation made the balls fly straighter than those fired from a smoothbore (non-rifled) barrel. Smoothbore sporting guns could fire a solid lead ball or, for shooting at birds, a measured quantity of small lead pellets, or "shot." In almost all cases, early muskets and rifles were muzzle-loaders, but they used a variety of ignition systems to fire the main charge. The guns shown here have matchlock (see p.22), wheel-lock (see pp.26–27), and flintlock (see pp.38–39) mechanisms. They have long barrels, which allows the gunpowder charge to burn fully, providing maximum power and greater accuracy.

▲ **GERMAN WHEEL-LOCK TSCHINKE**

Date	c.1630
Origin	Germany
Barrel	37in (94cm)
Caliber	.33in (8.3mm)

Wheel-locks exist in three basic forms: fully enclosed; with the wheel exposed but the rest of the lock enclosed; and with the entire mechanism exposed. The last form, known as a "Tschinke," a German wheel-lock, is more easily damaged but easier to clean and maintain. This example was made in Silesia (a region spanning areas of present-day Germany, Poland, and the Czech Republic), and its stock is inlaid with horn and mother-of-pearl. It has a short butt forming a "cheek" stock which is braced against the face instead of the shoulder when firing. The gun has a heavy barrel to help absorb much of the recoil when it fires.